

Growth Curve Analysis 2025

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```
library(here)
library(nlme)
library(ggplot2)
library(knitr)
library(dplyr)
library(effsize)
library(flextable)
library(officer)
# Note: MASS is used via MASS::mvrnorm() in the bootstrap functions
```

Prepare data

```
# Read full dataset including juveniles
growth_data_all <- read.csv(here("data", "processed", "growth_data.csv"),
                             na.strings = c("", ",", "NA", "."),
                             header = TRUE)

# Filter to adults (age >= 2) for von Bertalanffy models
# Age 2 is the approximate age of first reproduction in this species,
# so VB models are fitted to post-maturation growth only
growth_data <- growth_data_all %>%
  filter(!is.na(FSVLAFTER), !is.na(SITE.),
         !is.na(Age), !is.na(FEMID.),
         Age >= 2)

# Check sample sizes by age and site
table(growth_data$Age, growth_data$SITE.)
```

```
##
##      CP  OR
##  2   32  66
##  3  195 120
##  4  137  92
##  5   76  72
##  6   43  41
##  7   28  31
##  8   24  22
##  9   19  11
## 10    7   8
```

```
##    11    6    2
##    12    1    1
##    13    0    1
```

```
# Separate by site for site-specific model fitting
growth_data_OR <- growth_data %>% filter(SITE. == "OR")
growth_data_CP <- growth_data %>% filter(SITE. == "CP")
```

Summary statistics

```
summary_stats <- tibble(
  Population = c("Overall", "Warm lowland (OR)", "Cool highland (CP)"),
  Observations = c(nrow(growth_data), nrow(growth_data_OR), nrow(growth_data_CP)),
  Individuals = c(length(unique(growth_data$FEMID.)),
                  length(unique(growth_data_OR$FEMID.)),
                  length(unique(growth_data_CP$FEMID.))),
  Age_min = c(min(growth_data$Age), min(growth_data_OR$Age), min(growth_data_CP$Age)),
  Age_max = c(max(growth_data$Age), max(growth_data_OR$Age), max(growth_data_CP$Age)),
  Mean_obs_per_ind = c(
    round(mean(table(growth_data$FEMID.)), 1),
    round(mean(table(growth_data_OR$FEMID.)), 1),
    round(mean(table(growth_data_CP$FEMID.)), 1)),
  Obs_range_per_ind = c(
    paste0(min(table(growth_data$FEMID.)), "-", max(table(growth_data$FEMID.))),
    paste0(min(table(growth_data_OR$FEMID.)), "-", max(table(growth_data_OR$FEMID.))),
    paste0(min(table(growth_data_CP$FEMID.)), "-", max(table(growth_data_CP$FEMID.))))

kable(summary_stats, digits = 1,
       caption = "Summary statistics for adult growth data (age >= 2)")
```

Table 1: Summary statistics for adult growth data (age ≥ 2)

Population	Observations	Individuals	Age_min	Age_max	Mean_obs_per_ind	Obs_range_per_ind
Overall	1035	467	2	13	2.2	1-11
Warm lowland (OR)	467	206	2	13	2.3	1-11
Cool highland (CP)	568	276	2	12	2.1	1-9

Fit mixed-effects von Bertalanffy models

Von Bertalanffy growth models were fitted using nonlinear mixed effects (nlme), with individual identity (FEMID) as a random effect on the asymptotic size parameter (Linf). This accounts for repeated measurements on the same individual and allows individuals to vary in their asymptotic size while sharing common growth rate (K) and theoretical age at zero size (t0) parameters at the population level. Models fitted by maximum likelihood to allow AIC comparison.

```

# ORFORD (warm lowland population)
fit_vb_OR <- nlme(
  FSVLAFTER ~ Linf * (1 - exp(-K * (Age - t0))),
  data = growth_data_OR,
  fixed = Linf + K + t0 ~ 1,
  random = Linf ~ 1 | FEMID.,
  start = c(Linf = 74, K = 0.044, t0 = 0),
  control = nlmeControl(maxIter = 200, msMaxIter = 200))

summary(fit_vb_OR)

## Nonlinear mixed-effects model fit by maximum likelihood
## Model: FSVLAFTER ~ Linf * (1 - exp(-K * (Age - t0)))
## Data: growth_data_OR
##      AIC      BIC    logLik
## 2198.095 2218.827 -1094.048
##
## Random effects:
## Formula: Linf ~ 1 | FEMID.
##      Linf Residual
## StdDev: 2.808309 1.755631
##
## Fixed effects: Linf + K + t0 ~ 1
##      Value Std.Error DF t-value p-value
## Linf 68.66634 0.7181756 259 95.61219      0
## K      0.30807 0.0474663 259  6.49035      0
## t0    -4.68372 0.8883792 259 -5.27221      0
## Correlation:
## Linf K
## K -0.876
## t0 -0.804 0.987
##
## Standardized Within-Group Residuals:
##      Min      Q1      Med      Q3      Max
## -2.963959865 -0.496980541 -0.001496875  0.452527689  2.808763066
##
## Number of Observations: 467
## Number of Groups: 206

intervals(fit_vb_OR) # 95% CIs for fixed effects

## Approximate 95% confidence intervals
##
## Fixed effects:
##      lower      est.      upper
## Linf 67.2566851 68.6663420 70.0759988
## K      0.2149044 0.3080726 0.4012408
## t0    -6.4274571 -4.6837193 -2.9399816
##
## Random Effects:
## Level: FEMID.
##      lower      est.      upper

```

```
## sd(Linf) 2.45964 2.808309 3.206403
##
## Within-group standard error:
## lower est. upper
## 1.607184 1.755631 1.917790
```

```
# CENTRAL PLATEAU (cool highland population)
fit_vb_CP <- nlme(
  FSVLAFTER ~ Linf * (1 - exp(-K * (Age - t0))),
  data = growth_data_CP,
  fixed = Linf + K + t0 ~ 1,
  random = Linf ~ 1 | FEMID.,
  start = c(Linf = 89, K = 0.044, t0 = 0),
  control = nlmeControl(maxIter = 200, msMaxIter = 200))

summary(fit_vb_CP)
```

```
## Nonlinear mixed-effects model fit by maximum likelihood
## Model: FSVLAFTER ~ Linf * (1 - exp(-K * (Age - t0)))
## Data: growth_data_CP
## AIC BIC logLik
## 2828.76 2850.47 -1409.38
##
## Random effects:
## Formula: Linf ~ 1 | FEMID.
## Linf Residual
## StdDev: 3.209841 1.975644
##
## Fixed effects: Linf + K + t0 ~ 1
## Value Std.Error DF t-value p-value
## Linf 81.08730 0.5558443 290 145.88132 0
## K 0.41218 0.0384590 290 10.71739 0
## t0 -2.34545 0.3969407 290 -5.90883 0
## Correlation:
## Linf K
## K -0.791
## t0 -0.681 0.980
##
## Standardized Within-Group Residuals:
## Min Q1 Med Q3 Max
## -3.40104709 -0.42009765 -0.02162272 0.40815809 4.13171224
##
## Number of Observations: 568
## Number of Groups: 276
```

```
intervals(fit_vb_CP)
```

```
## Approximate 95% confidence intervals
##
## Fixed effects:
## lower est. upper
## Linf 79.9961890 81.0872966 82.1784041
## K 0.3366862 0.4121803 0.4876743
```

```
## t0    -3.1246378 -2.3454537 -1.5662697
##
## Random Effects:
## Level: FEMID.
##          lower      est.      upper
## sd(Linf) 2.854841 3.209841 3.608985
##
## Within-group standard error:
##      lower      est.      upper
## 1.816707 1.975644 2.148486
```

Model diagnostics

```
# Function to produce standard diagnostic plots for nlme models
check_nlme_diagnostics <- function(model, title) {

  par(mfrow = c(2, 2))

  # 1. Residuals vs Fitted - checks for systematic patterns (non-linearity, bias)
  plot(fitted(model), residuals(model, type = "pearson"),
       xlab = "Fitted values", ylab = "Pearson residuals",
       main = paste(title, "- Residuals vs Fitted"))
  abline(h = 0, col = "red", lty = 2)
  lines(lowess(fitted(model), residuals(model, type = "pearson")),
        col = "blue", lwd = 2)

  # 2. QQ plot - checks normality of residuals
  qqnorm(residuals(model, type = "pearson"),
         main = paste(title, "- Normal Q-Q"))
  qqline(residuals(model, type = "pearson"), col = "red")

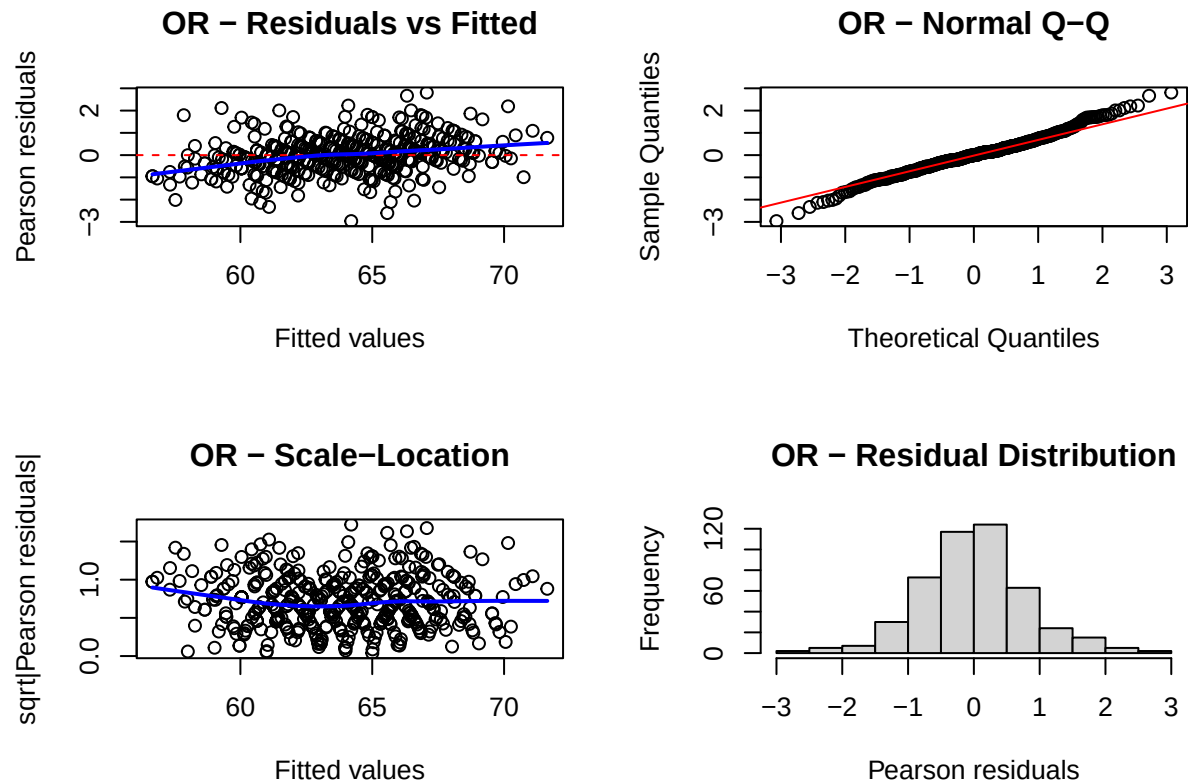
  # 3. Scale-Location - checks homoscedasticity (constant variance)
  plot(fitted(model), sqrt(abs(residuals(model, type = "pearson"))),
       xlab = "Fitted values", ylab = "sqrt|Pearson residuals|",
       main = paste(title, "- Scale-Location"))
  lines(lowess(fitted(model), sqrt(abs(residuals(model, type = "pearson")))),
        col = "blue", lwd = 2)

  # 4. Histogram - visualises residual distribution
  hist(residuals(model, type = "pearson"), breaks = 20,
       main = paste(title, "- Residual Distribution"),
       xlab = "Pearson residuals")

  par(mfrow = c(1, 1))

  # Shapiro-Wilk test for normality of residuals
  # Note: at large sample sizes (n > 400) this test is highly sensitive and
  # significant results should be interpreted alongside visual diagnostics
  cat("\n", title, "- Shapiro-Wilk test for normality:\n")
  print(shapiro.test(residuals(model, type = "pearson")))
}
```

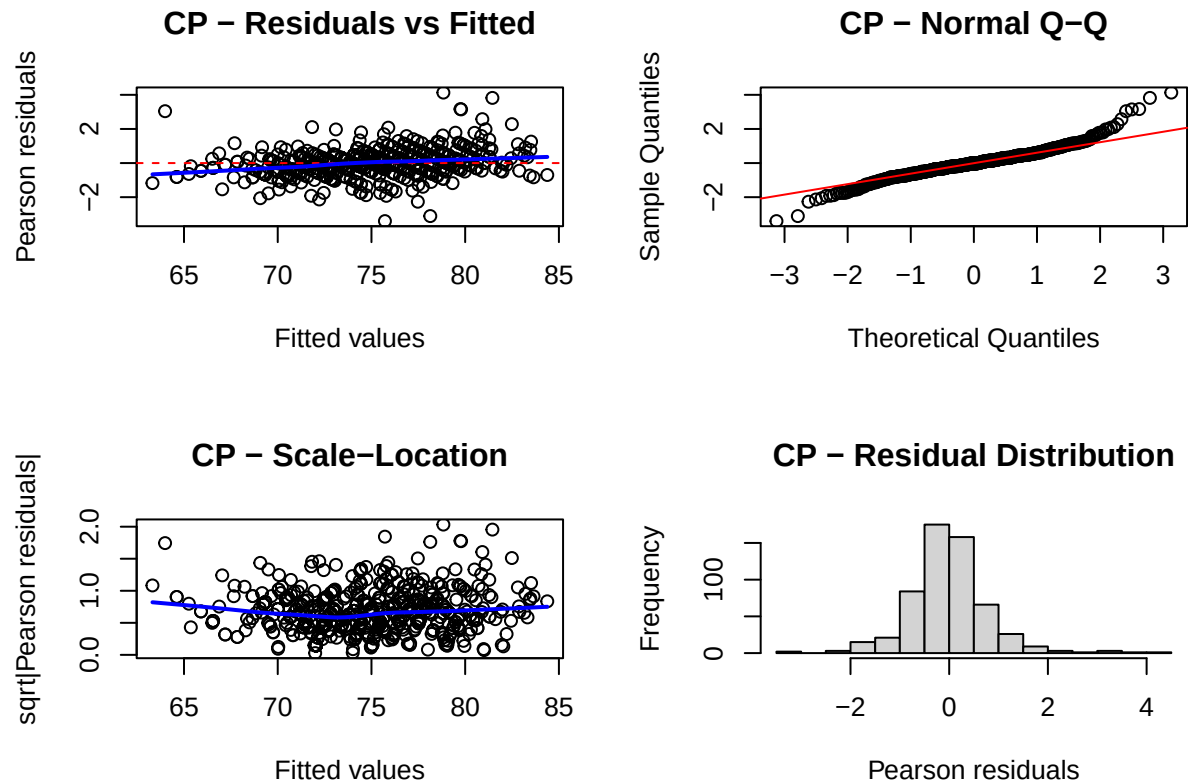
```
check_nlme_diagnostics(fit_vb_OR, "OR")
```



```
##
## OR - Shapiro-Wilk test for normality:
##
## Shapiro-Wilk normality test
##
## data: residuals(model, type = "pearson")
## W = 0.98874, p-value = 0.001162
```

```
# OR diagnostics: Shapiro-Wilk is significant (p = 0.001) but visual inspection
# of the QQ plot shows only minor deviation at the tails, with residuals
# following the theoretical line closely across the bulk of the distribution.
# Given the large sample size (n = 467), this level of Shapiro-Wilk sensitivity
# is expected and does not indicate a meaningful violation. Residuals vs fitted
# and scale-location plots show no systematic patterns.
```

```
check_nlme_diagnostics(fit_vb_CP, "CP")
```

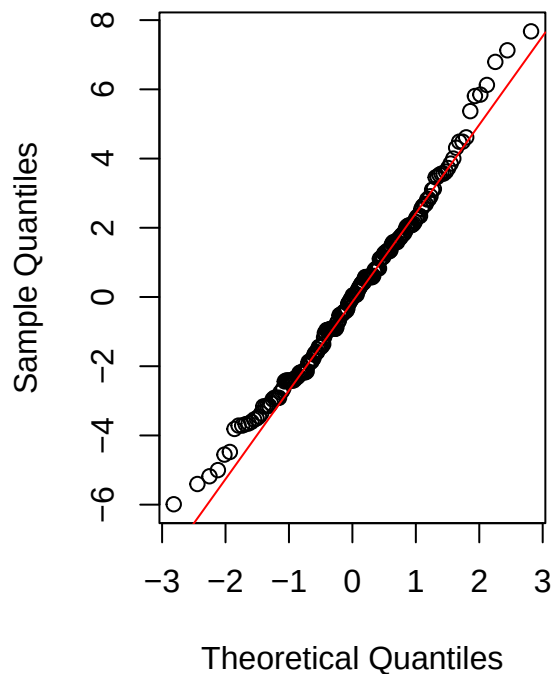


```
##
## CP - Shapiro-Wilk test for normality:
##
## Shapiro-Wilk normality test
##
## data: residuals(model, type = "pearson")
## W = 0.94777, p-value = 2.858e-13
```

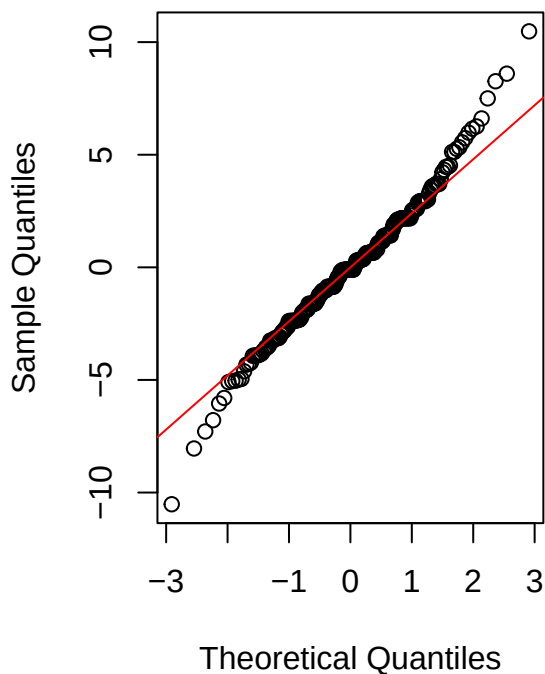
```
# CP diagnostics: Both Shapiro-Wilk (p = 2.9e-13) and visual inspection indicate
# more substantial departure from normality, with heavy tails on the QQ plot and
# a possible outlier at ~4 Pearson residuals. The scale-location plot suggests
# mild heteroscedasticity (increasing variance with fitted values).
# Testing variance structures below to assess whether this can be improved.
```

```
# Check normality of random effects (individual-level Linf deviations)
# These should also be approximately normal for valid inference
par(mfrow = c(1, 2))
qqnorm(ranef(fit_vb_OR)[,1], main = "OR - Random Effects Q-Q")
qqline(ranef(fit_vb_OR)[,1], col = "red")
qqnorm(ranef(fit_vb_CP)[,1], main = "CP - Random Effects Q-Q")
qqline(ranef(fit_vb_CP)[,1], col = "red")
```

OR – Random Effects Q-Q



CP – Random Effects Q-Q



```
par(mfrow = c(1, 1))
# Random effects are approximately normally distributed for both populations.
```

Test variance structures for CP heteroscedasticity

The CP model showed mild heteroscedasticity in residual diagnostics. Two variance structures were tested to assess whether explicit variance modelling could improve fit: varPower and varExp (both as functions of fitted values).

```
# Model 2: variance as power function of fitted values
fit_vb_CP_power <- nlme(
  FSVLAFTER ~ Linf * (1 - exp(-K * (Age - t0))),
  data = growth_data_CP,
  fixed = Linf + K + t0 ~ 1,
  random = Linf ~ 1 | FEMID.,
  weights = varPower(form = ~fitted()),
  start = c(Linf = 89, K = 0.044, t0 = 0),
  control = nlmeControl(maxIter = 200, msMaxIter = 200))
```

```
## Warning in nlme.formula(FSVLAFTER ~ Linf * (1 - exp(-K * (Age - t0))), data =
## growth_data_CP, : Iteration 7, LME step: nlminb() did not converge (code = 1).
## PORT message: false convergence (8)
```

```

# Note: model produced a convergence warning during fitting

# Model 3 (varExp) failed to converge entirely (maxIter = 200 reached),
# indicating the exponential variance structure is not supported by these data.

# Compare base model vs power variance model
AIC(fit_vb_CP, fit_vb_CP_power)

```

```

##           df      AIC
## fit_vb_CP      5 2828.760
## fit_vb_CP_power 6 2823.564

```

```

summary(fit_vb_CP_power)

```

```

## Nonlinear mixed-effects model fit by maximum likelihood
## Model: FSVLAFTER ~ Linf * (1 - exp(-K * (Age - t0)))
## Data: growth_data_CP
##      AIC      BIC    logLik
## 2823.564 2849.616 -1405.782
##
## Random effects:
## Formula: Linf ~ 1 | FEMID.
##           Linf      Residual
## StdDev: 3.195326 1.589092e-07
##
## Variance function:
## Structure: Power of variance covariate
## Formula: ~fitted(.)
## Parameter estimates:
##      power
## 3.778684
## Fixed effects: Linf + K + t0 ~ 1
##           Value Std.Error DF   t-value p-value
## Linf 80.63911 0.5439287 290 148.25309      0
## K      0.43582 0.0390727 290 11.15420      0
## t0    -2.12497 0.3558022 290 -5.97233      0
## Correlation:
##      Linf      K
## K    -0.797
## t0   -0.689  0.980
##
## Standardized Within-Group Residuals:
##           Min           Q1           Med           Q3           Max
## -4.264598218 -0.366454484  0.002544275  0.447492605  3.990596685
##
## Number of Observations: 568
## Number of Groups: 276

```

```

# Assessment: despite deltaAIC ~ 5 in favour of varPower, the model produced
# a degenerate solution - residual SD collapsed to ~0 (1.59e-07) with an
# implausibly large power parameter (~3.78), indicating the variance function
# absorbed all residual variance rather than capturing meaningful heteroscedasticity.

```

```

# The convergence warning and complete failure of varExp to converge further
# indicate this heteroscedasticity is not sufficiently structured to support
# explicit variance modelling.
# Base model retained; mild heteroscedasticity noted as a limitation.
# Fixed effect estimates are unlikely to be materially biased (n = 568).
fit_vb_CP_final <- fit_vb_CP

```

Model selection: compare VB, logistic, and Gompertz

Three commonly used growth functions were compared by AIC to confirm VB as the most appropriate functional form for each population.

```

# Logistic growth: FSVLAFTER = Linf / (1 + exp(-K * (Age - t0)))
fit_log_OR <- nlme(
  FSVLAFTER ~ Linf / (1 + exp(-K * (Age - t0))),
  data = growth_data_OR,
  fixed = Linf + K + t0 ~ 1,
  random = Linf ~ 1 | FEMID.,
  start = c(Linf = 74, K = 0.5, t0 = 1),
  control = nlmeControl(maxIter = 200, msMaxIter = 200))

fit_log_CP <- nlme(
  FSVLAFTER ~ Linf / (1 + exp(-K * (Age - t0))),
  data = growth_data_CP,
  fixed = Linf + K + t0 ~ 1,
  random = Linf ~ 1 | FEMID.,
  start = c(Linf = 89, K = 0.5, t0 = 1),
  control = nlmeControl(maxIter = 200, msMaxIter = 200))

# Gompertz growth: FSVLAFTER = Linf * exp(-exp(-K * (Age - t0)))
fit_gomp_OR <- nlme(
  FSVLAFTER ~ Linf * exp(-exp(-K * (Age - t0))),
  data = growth_data_OR,
  fixed = Linf + K + t0 ~ 1,
  random = Linf ~ 1 | FEMID.,
  start = c(Linf = 74, K = 0.5, t0 = 0),
  control = nlmeControl(maxIter = 200, msMaxIter = 200))

fit_gomp_CP <- nlme(
  FSVLAFTER ~ Linf * exp(-exp(-K * (Age - t0))),
  data = growth_data_CP,
  fixed = Linf + K + t0 ~ 1,
  random = Linf ~ 1 | FEMID.,
  start = c(Linf = 89, K = 0.5, t0 = 0),
  control = nlmeControl(maxIter = 200, msMaxIter = 200))

# AIC comparison within each population
AIC(fit_vb_OR, fit_log_OR, fit_gomp_OR)

```

```

##          df      AIC
## fit_vb_OR    5 2198.095
## fit_log_OR    5 2198.419

```

```
## fit_gomp_OR 5 2198.225
```

```
AIC(fit_vb_CP, fit_log_CP, fit_gomp_CP)
```

```
##          df      AIC
## fit_vb_CP    5 2828.760
## fit_log_CP    5 2831.054
## fit_gomp_CP    5 2829.900
```

```
# All three models have deltaAIC < 2 within each population, indicating
# they are statistically indistinguishable in fit. VB was selected as the
# preferred model on the basis of biological convention (widely used for
# reptile growth) and interpretability of parameters (Linf, K, t0),
# not statistical superiority.
```

Site comparison: combined model

A single combined model with site-specific fixed effects for all three parameters was used to formally test whether growth parameters differed between populations. CP is the reference level (alphabetical ordering), so intercepts represent CP estimates and SiteOR coefficients represent OR deviations from CP.

```
growth_data_both <- growth_data %>%
  mutate(Site = factor(SITE.))
# Note: factor() will set CP as reference level (alphabetical order)
# Intercepts = CP estimates; SiteOR coefficients = difference (OR - CP)

fit_combined <- nlme(
  FSVLAFTER ~ Linf * (1 - exp(-K * (Age - t0))),
  data = growth_data_both,
  fixed = list(Linf ~ Site, K ~ Site, t0 ~ Site),
  random = Linf ~ 1 | FEMID.,
  start = c(81, -12, 0.41, -0.13, -2.35, -2.83),
  # start values: Linf_CP, Linf diff(OR-CP), K_CP, K diff(OR-CP), t0_CP, t0 diff(OR-CP)
  control = nlmeControl(maxIter = 200, msMaxIter = 200))

summary(fit_combined)
```

```
## Nonlinear mixed-effects model fit by maximum likelihood
## Model: FSVLAFTER ~ Linf * (1 - exp(-K * (Age - t0)))
## Data: growth_data_both
##      AIC      BIC    logLik
## 5058.766 5098.303 -2521.383
##
## Random effects:
## Formula: Linf ~ 1 | FEMID.
##      Linf.(Intercept) Residual
## StdDev:      2.96705 1.959625
##
## Fixed effects: list(Linf ~ Site, K ~ Site, t0 ~ Site)
##      Value Std.Error DF  t-value p-value
## Linf.(Intercept) 80.79018 0.5348841 563 151.04238 0.0000
```

```
## Linf.SiteOR      -11.88579  1.0108060 563 -11.75873  0.0000
## K.(Intercept)    0.41551  0.0385392 563  10.78137  0.0000
## K.SiteOR         -0.12877  0.0648676 563  -1.98509  0.0476
## t0.(Intercept)   -2.35443  0.3954349 563  -5.95402  0.0000
## t0.SiteOR        -2.83106  1.1783812 563  -2.40250  0.0166
## Correlation:
##               Ln.(I) Ln.SOR K.(In) K.StOR t0.(I)
## Linf.SiteOR      -0.525
## K.(Intercept)    -0.802  0.430
## K.SiteOR          0.477 -0.865 -0.599
## t0.(Intercept)   -0.694  0.373  0.980 -0.587
## t0.SiteOR         0.232 -0.785 -0.335  0.944 -0.342
##
## Standardized Within-Group Residuals:
##           Min           Q1           Med           Q3           Max
## -3.42672837 -0.44567282 -0.01040621  0.41807746  4.20571366
##
## Number of Observations: 1035
## Number of Groups: 467
```

```
# Sequential F-tests for significance of site differences in each parameter
anova(fit_combined)
```

```
##               numDF denDF  F-value p-value
## Linf.(Intercept)      1   563 400550.0 <.0001
## Linf.Site              1   563  1772.0 <.0001
## K.(Intercept)          1   563 11024.5 <.0001
## K.Site                 1   563   100.4 <.0001
## t0.(Intercept)         1   563    52.0 <.0001
## t0.Site                1   563     5.8  0.0166
```

```
# CP has significantly larger asymptotic size (Linf) and faster growth rate (K) than OR.
# t0 also differs significantly between sites, though the effect is smaller.
# OR approaches its (lower) asymptote more slowly than CP approaches its (larger) asymptote.
```

Results table

```
# --- Table 1: VB parameters with 95% CIs ---
results_table <- tibble(
  Population = rep(c("Warm lowland (OR)", "Cool highland (CP)"), each = 3),
  Parameter = rep(c("Linf (mm)", "K (year-1)", "t0 (year)"), 2),
  Estimate = c(fixef(fit_vb_OR), fixef(fit_vb_CP_final)),
  Lower_95CI = c(intervals(fit_vb_OR)$fixed[, "lower"],
                 intervals(fit_vb_CP_final)$fixed[, "lower"]),
  Upper_95CI = c(intervals(fit_vb_OR)$fixed[, "upper"],
                 intervals(fit_vb_CP_final)$fixed[, "upper"])) %>%
  mutate(across(c(Estimate, Lower_95CI, Upper_95CI), ~round(., 3)))

# --- Table 2: ANOVA results for site differences ---
anova_results <- as.data.frame(anova(fit_combined)) %>%
  tibble::rownames_to_column("Term") %>%
```

```

filter(grepl("Site", Term)) %>%
mutate(
  Parameter = c("Linf (mm)", "K (year-1)", "t0 (year)"),
  F_value = round(`F-value`, 1),
  p_value = c("<0.001", "<0.001", "0.017")
) %>%
dplyr::select(Parameter, numDF, denDF, F_value, p_value)

# --- Save both tables to Word ---
ft1 <- flextable(results_table) %>%
  set_caption("Von Bertalanffy growth parameters and 95% CIs for each population")

ft2 <- flextable(anova_results) %>%
  set_caption("Population differences in von Bertalanffy growth parameters (F-tests from combined model)")

doc <- read_docx() %>%
  body_add_par("Table 1: Growth parameters", style = "heading 1") %>%
  body_add_flextable(ft1) %>%
  body_add_par("") %>%
  body_add_par("Table 2: Population comparisons", style = "heading 1") %>%
  body_add_flextable(ft2)

print(doc, target = here("outputs", "tables", "Supp_growth.docx"))

```

Early growth comparison (ages 0-2 years)

To determine whether population differences in growth are present from birth and through the pre-reproductive phase, SVL was compared between populations at age 0.01 years (all offspring born in the lab are measured at birth, $n = 1660$ OR and 2250 CP) and a subset of juveniles that were recaptured for survival analysis in Spring following their first winter dormant period (approximate age 0.75, $n = 187$).

Only these two discrete age classes are available prior to adulthood reflecting the measurement schedule of the long-term study. A linear mixed effects model would be preferable to account for individual identity, but of the offspring born (age 0.01, only approx 20% survive to age 2, so we cannot include ID). A linear model fit to two data points is unsuitable therefore we use a simple t-test between the two populations at the two juvenile age classes. As the sample size at birth is very large ($n > 4000$ across age classes), even small differences will be statistically significant, and effect sizes should be interpreted alongside the model output.

```

# --- Split into two age classes ---

birth_data <- growth_data_all %>%
  filter(Age == 0.01) %>%
  mutate(Site = factor(SITE., levels = c("OR", "CP")))

juvenile_data <- growth_data_all %>%
  filter(Age == 0.75) %>%
  mutate(Site = factor(SITE., levels = c("OR", "CP")))

# --- Check sample sizes ---
cat("Birth sample sizes:\n")

```

```
## Birth sample sizes:
```

```
table(birth_data$Site)
```

```
##  
##   OR   CP  
## 1660 2267
```

```
cat("\nJuvenile sample sizes:\n")
```

```
##  
## Juvenile sample sizes:
```

```
table(juvenile_data$Site)
```

```
##  
##   OR   CP  
##   57  130
```

```
# --- Descriptive statistics ---
```

```
birth_data %>%  
  group_by(Site) %>%  
  summarise(n = n(),  
            mean_SVL = mean(FSVLAFTER, na.rm = TRUE),  
            sd_SVL = sd(FSVLAFTER, na.rm = TRUE),  
            se_SVL = sd_SVL / sqrt(n))
```

```
## # A tibble: 2 x 5  
##   Site      n mean_SVL sd_SVL se_SVL  
##   <fct> <int>   <dbl> <dbl> <dbl>  
## 1 OR      1660    29.6  0.894 0.0220  
## 2 CP      2267    30.2  0.925 0.0194
```

```
juvenile_data %>%  
  group_by(Site) %>%  
  summarise(n = n(),  
            mean_SVL = mean(FSVLAFTER, na.rm = TRUE),  
            sd_SVL = sd(FSVLAFTER, na.rm = TRUE),  
            se_SVL = sd_SVL / sqrt(n))
```

```
## # A tibble: 2 x 5  
##   Site      n mean_SVL sd_SVL se_SVL  
##   <fct> <int>   <dbl> <dbl> <dbl>  
## 1 OR      57     37.3   2.04 0.270  
## 2 CP     130     38.3   3.94 0.345
```

```
# --- Welch's t-tests ---
```

```
t_birth <- t.test(FSVLAFTER ~ Site, data = birth_data, var.equal = FALSE)  
t_juvenile <- t.test(FSVLAFTER ~ Site, data = juvenile_data, var.equal = FALSE)  
  
print(t_birth)
```

```
##
## Welch Two Sample t-test
##
## data: FSVLAFTER by Site
## t = -19.463, df = 3621.2, p-value < 2.2e-16
## alternative hypothesis: true difference in means between group OR and group CP is not equal to 0
## 95 percent confidence interval:
## -0.6301842 -0.5148414
## sample estimates:
## mean in group OR mean in group CP
## 29.63075 30.20326
```

```
print(t_juvenile)
```

```
##
## Welch Two Sample t-test
##
## data: FSVLAFTER by Site
## t = -2.2975, df = 179.92, p-value = 0.02274
## alternative hypothesis: true difference in means between group OR and group CP is not equal to 0
## 95 percent confidence interval:
## -1.8722874 -0.1421661
## sample estimates:
## mean in group OR mean in group CP
## 37.26316 38.27038
```

```
# --- Effect sizes (Cohen's d) ---
```

```
d_birth <- cohen.d(FSVLAFTER ~ Site, data = birth_data)
d_juvenile <- cohen.d(FSVLAFTER ~ Site, data = juvenile_data)

print(d_birth)
```

```
##
## Cohen's d
##
## d estimate: -0.627432 (medium)
## 95 percent confidence interval:
## lower upper
## -0.6924744 -0.5623895
```

```
print(d_juvenile)
```

```
##
## Cohen's d
##
## d estimate: -0.2899977 (small)
## 95 percent confidence interval:
## lower upper
## -0.60479898 0.02480351
```

```
# --- Summary table ---
results_early_growth <- data.frame(
  Age_class = c("Birth (~0.01 years)", "Post-winter (~0.75 years)"),
  n_OR = c(nrow(filter(birth_data, Site == "OR")),
           nrow(filter(juvenile_data, Site == "OR"))),
  n_CP = c(nrow(filter(birth_data, Site == "CP")),
           nrow(filter(juvenile_data, Site == "CP"))),
  mean_OR = c(mean(filter(birth_data, Site == "OR")$FSVLATER),
               mean(filter(juvenile_data, Site == "OR")$FSVLATER)),
  mean_CP = c(mean(filter(birth_data, Site == "CP")$FSVLATER),
               mean(filter(juvenile_data, Site == "CP")$FSVLATER)),
  t_stat = c(t_birth$statistic, t_juvenile$statistic),
  df = c(t_birth$parameter, t_juvenile$parameter),
  p_value = c(t_birth$p.value, t_juvenile$p.value),
  cohens_d = c(d_birth$estimate, d_juvenile$estimate)
) %>%
  mutate(across(where(is.numeric), ~round(., 3)))

print(results_early_growth)
```

```
##           Age_class n_OR n_CP mean_OR mean_CP t_stat      df p_value
## 1      Birth (~0.01 years) 1660 2267      NA      NA -19.463 3621.157  0.000
## 2 Post-winter (~0.75 years)   57  130  37.263   38.27  -2.298  179.922  0.023
## cohens_d
## 1    -0.627
## 2    -0.290
```

For publication plot see Figure 2 script Panel A

Session information

```
sessionInfo()
```

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26200)
##
## Matrix products: default
## LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=English_Australia.utf8 LC_CTYPE=English_Australia.utf8
## [3] LC_MONETARY=English_Australia.utf8 LC_NUMERIC=C
## [5] LC_TIME=English_Australia.utf8
##
## time zone: Australia/Hobart
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
```

```

##
## other attached packages:
## [1] officer_0.7.3    flextable_0.9.11 effsize_0.8.1    dplyr_1.1.4
## [5] knitr_1.51       ggplot2_4.0.2    nlme_3.1-168      here_1.0.2
##
## loaded via a namespace (and not attached):
## [1] utf8_1.2.6        generics_0.1.4    tidyr_1.3.2
## [4] fontLiberation_0.1.0 xml2_1.5.2        lattice_0.22-7
## [7] digest_0.6.39     magrittr_2.0.3    evaluate_1.0.5
## [10] grid_4.5.1        RColorBrewer_1.1-3 fastmap_1.2.0
## [13] rprojroot_2.1.1   zip_2.3.3         purrr_1.2.1
## [16] scales_1.4.0      fontBitstreamVera_0.1.1 textshaping_1.0.4
## [19] cli_3.6.5         rlang_1.1.6       fontquiver_0.2.1
## [22] withr_3.0.2       yaml_2.3.12       otel_0.2.0
## [25] gdtools_0.5.0     tools_4.5.1       uuid_1.2-2
## [28] vctr_0.6.5        R6_2.6.1          lifecycle_1.0.5
## [31] ragg_1.5.0        pkgconfig_2.0.3   pillar_1.11.1
## [34] gtable_0.3.6      data.table_1.18.2.1 glue_1.8.0
## [37] Rcpp_1.1.1        systemfonts_1.3.1 xfun_0.56
## [40] tibble_3.3.0      tidyselect_1.2.1  rstudioapi_0.18.0
## [43] farver_2.1.2      patchwork_1.3.2   htmltools_0.5.9
## [46] rmarkdown_2.30    compiler_4.5.1    S7_0.2.0
## [49] askpass_1.2.1     openssl_2.3.4

```